FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – IX

PAPER-1

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**B, C, D**	19.	**D**	37.	**B, C**
2.	**A, B, C, D**	20.	**A**	38.	**B, D**
3.	**C**	21.	**D**	39.	**B, D**
4.	**A, D**	22.	**A, D**	40.	**B, C**
5.	**A, D**	23.	**B, C**	41.	**A, C**
6.	**A, B, C, D**	24.	**A, B, D**	42.	**B, D**
7.	**B, D**	25.	**A, B, D**	43.	**A, C**
8.	**C**	26.	**B**	44.	**C**
9.	**A**	27.	**A**	45.	**B**
10.	**D**	28.	**C**	46.	**C**
11.	**D**	29.	**C**	47.	**C**
12.	**A**	30.	**C**	48.	**C**
13.	**C**	31.	**C**	49.	**C**
14.	**4**	32.	**4**	50.	**3**
15.	**5**	33.	**2**	51.	**4**
16.	**6**	34.	**4**	52.	**3**
17.	**6**	35.	**5**	53.	**3**
18.	**3**	36.	**5**	54.	**5**

Physics

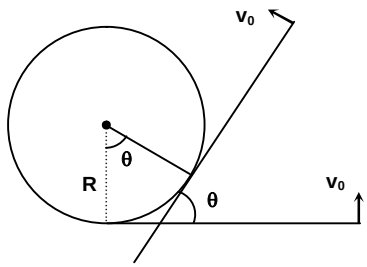
PART – I

SECTION – A

1. In order to flow up the floor drain, the KE of the fluid must be just sufficient to provide for the gravitational P.E. required:
$v_M^2 = 2gh_2$
The flow rate through M is :
$V = \left(\pi r_M^2\right) v_M;\ r_M = 3\,cm = 0.03\,m$
and this is the total discharge from the roof.
The rainfall rate is $\frac{V}{wL}$ (in m/s) and it can be converted to cm/h by multiplying by (100×3600).

2. All are based on true facts.

3. T and v are perpendicular so speed is constant.



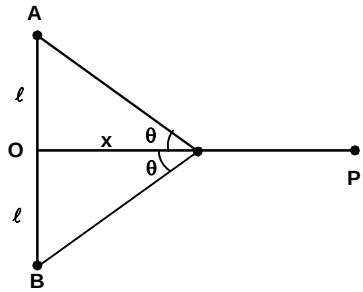
$\theta = \frac{\ell - x}{R}$
$\Rightarrow \frac{d\theta}{dt} = -\frac{dx}{Rdt}$
$\Rightarrow \frac{v_0}{x} = -\frac{dx}{Rdt}$
$\Rightarrow xdx = -Rv_0dt$
$\Rightarrow t = \frac{\ell^2}{2Rv_0}$
Now check the options.

4. Use horizontal & vertical velocity components for comparison.

5. Kinetic energy can never be negative.

6. Use of coulomb forces.

7. Net B at distance x from O along x-axis



$= \frac{\mu_0}{2\pi}\frac{I}{r}2\cos\theta$
$\Rightarrow \int dF = \int I\frac{\mu_0}{2\pi}\frac{I}{r}2\cos\theta dx$
$= \frac{\mu_0}{2\pi}I^2\ell n\left(1+\frac{L^2}{\ell^2}\right)$

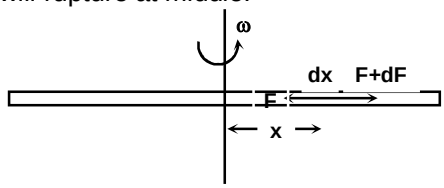
8-10. Along the incline plane net torque on the cylinder should be zero.
For emf $e = \frac{d\phi}{dt}$ and $\frac{de}{dt} = \frac{d}{dt}\left(\frac{d\phi}{dt}\right)$

11-13. In first column there are some laws and in second column associated formulae. By using these formulae solve the problems.

SECTION – C

14. at t = 2
$y = 2 \times 2 = 4$
so $y = x^2$
$\Rightarrow \; 4 = x^2$
$\Rightarrow \; x = 2$
$\Rightarrow$ emf $e = vB\ell$
$= 2 \times 0.5 \times 4 = 4$ Volt

15. Stress are zero at free ends and maximum at middle so rod will rupture at middle.
By Newton's second law



$F - (F + dF) = dmx\omega^2$
$-\int dF = \int \rho A\omega^2 \times dx$
Where ω is the speed of rotation

$$-[F]_0^F = \rho A\omega^2 \left[\frac{x^2}{z}\right]_{\ell/2}^{x}$$

$$+F = +\rho\frac{A\omega^2}{z}\left[\left(\frac{\ell}{2}\right)^2 - x^2\right]$$

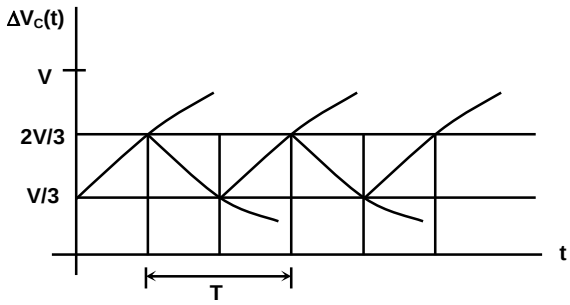
At $x = 0$ $F = \dfrac{\rho A\omega^2\ell^2}{8}$

$$\Rightarrow \quad \frac{F}{A} = \frac{\rho\omega^2\ell^2}{8}$$

For rupture $\dfrac{F}{A} = \sigma \;\Rightarrow\; \dfrac{\rho\omega^2\ell^2}{8} = \sigma$

$$\Rightarrow \quad n = \frac{1}{2\pi}\sqrt{\frac{8\sigma}{\rho\ell^2}}$$

16. Start with $\Delta V_C = \dfrac{2V}{3}$; By discharging formula



$$\Delta V_C(t) = \frac{2}{3}Ve^{-t_1/R_2C}$$

For $\Delta V_C(t) = \dfrac{V}{3}$ we get $t_1 = R_2C\ln 2$

After the switch opens voltage is $\dfrac{V}{3}$; increasing towards V with time constant $(R_1 + R_2)C$ so

$$\Delta V_C(t) = \Delta V - \frac{2}{3}\Delta V e^{-t_2/(R_1+R_2)C}$$

$$\Rightarrow \quad t_2 = (R_1 + R_2)C\ln 2$$

17. The current through E_2 is given by the solid line, while that through E_1 is given by the dotted line. We can choose points on the graph, apply KVL and KCL to the circuit and substitute these values to determine the unknowns. This gives $E_1 = 6\,V$.

18. Acceleration of system = $\frac{F}{M+m}$

Tension at distance x from A at t = t

$T = m'a = \frac{Fmx}{\ell(M+m)}$

$\Rightarrow$ Speed of pulse at x distance from A is

$v = \sqrt{\frac{Fmx}{\ell(M+m)}\frac{\ell}{m}} = \sqrt{\frac{Fx}{M+m}}$

$\Rightarrow \quad a = \frac{F}{2(M+m)}$

Chemistry PART – II

SECTION – A

19. Second excited state of H^- is 2p, which has one angular node.

20. $\alpha = \sqrt{\frac{K}{C}}$, greater be the value of K, stronger be the electrolyte.

21. Greater be the number of covalent bonds, greater be the stability.

22. For solid and liquid $C_P \approx C_V$.
Below critical point, gases may be liquefied through the application of pressure.
Thus, $z < 1$.

23. Si cannot form effective $p\pi$ - $p\pi$ bond with oxygen due to larger size of 3p orbital of Si. In borax, both sp^2 and sp^3 hybridised 'B' atoms are present.

24. Due to presence of unpaired electron.

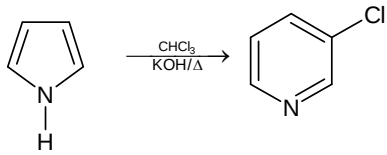
25. In the haloform reaction, extraction of proton is RDS. The reaction does not depend on nature and concentration of halogen.

26. - OH group of phenol is strongly activating and bromination in aqueous medium gives tribromo derivative.

27. - COOH is strongly deactivating and formed by oxidation of toluene.

28. $-CH_3$ is weakly activating.

SECTION – C

32. 

33. $XeF_6 + H_2O \longrightarrow XeOF_4 + 2HF(\text{partial hydrolysis})$
$XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF(\text{complete hydrolysis})$
Number of lone pair in $XeOF_4$ = 1
Number of lone pair in XeO_3 = 1
Total number of lone pairs = 2.

Mathematics PART – III

SECTION – A

37. $f(x) = 1 - \dfrac{x+1}{x^2 - 5x + 6}$

$f'(x) > 0 \Rightarrow x^2 + 2x - 11 > 0$

$x > 2\sqrt{3} - 1$ or $x < -2\sqrt{3} - 1$

$f \uparrow$ in $\left(-\infty, -2\sqrt{3} - 1\right) \cup \left(2\sqrt{3} - 1, 3\right) \cup (3, \infty)$

$f \downarrow$ in $\left(-2\sqrt{3} - 1, 2\right) \cup \left(2, 2\sqrt{3} - 1\right)$

$f\left(2\sqrt{3} - 1\right) = 8 + 4\sqrt{3}$

$f\left(-2\sqrt{3} - 1\right) = 8 - 4\sqrt{3}$

$\lim\limits_{x \to -\infty} f(x) = \lim\limits_{x \to \infty} f(x) = 1$

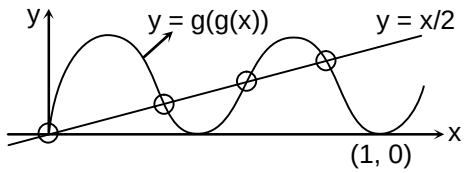
38. $\displaystyle\int_{-1}^{1} f(x+1)\,dx = \int_{-1}^{0} f(x+1)\,dx + \int_{0}^{1} f(x+1)\,dx = \int_{0}^{1} \left(f(x) + f(x+1)\right)dx = \int_{0}^{1} \left(x + \{x\}\right)dx = 1$

$\displaystyle I = \int_{1/2}^{3/2} g\left(g(x)\right)dx = \int_{1/2}^{3/2} g\left(g(2-x)\right)dx = \int_{1/2}^{3/2} g\left(2 - g(x)\right)dx = \int_{1/2}^{3/2} \left(2 - g\left(g(x)\right)\right)dx$

$\displaystyle I = \int_{1/2}^{3/2} dx = \frac{3}{2} - \frac{1}{2} = 1$

39. Now $g(x) = 0 \Rightarrow x = 0, 1$ also, $g\left(\dfrac{1}{2}\right) = 1$

So, $g(g(0)) = g(g(1)) = g\left(g\left(\dfrac{1}{2}\right)\right) = 0 \Rightarrow n_1 = 3$



Also, $g'(g(x))g'(x) = 0$

$\Rightarrow x = \dfrac{1}{2}$, $g(x) = \dfrac{1}{2} \Rightarrow x = \dfrac{1}{2} \pm \dfrac{1}{2\sqrt{2}}$

$gg\left(\dfrac{1}{2} + \dfrac{1}{2\sqrt{2}}\right) = 1$

So, the number of solution is $4 \Rightarrow n_2 = 4$

40. By properties $PA = PB = \sqrt{10}$

$\dfrac{x-1}{-1/\sqrt{10}} = \dfrac{y-5}{3/\sqrt{10}} = \pm\sqrt{10}$

$\Rightarrow A \equiv (0, 8)$; $B \equiv (2, 2)$

Slopes of asymptotes are –7, and 1

If angle between asymptotes be θ then $\tan\theta = \left|\dfrac{-7-1}{1-7}\right| = \dfrac{4}{3}$

Now, $2\tan^{-1}\dfrac{b}{a} = \tan^{-1}\dfrac{4}{3} \Rightarrow \dfrac{\dfrac{2b}{a}}{1 - \dfrac{b^2}{a^2}} = \dfrac{4}{3}$

$\Rightarrow \frac{b}{a} = \frac{1}{2}$ ..... (1)

$\frac{b^2}{a^2} = e^2 - 1 \Rightarrow e = \frac{\sqrt{5}}{2}$

Also, area of $\Delta CAB = ab$

$ab = \left|\frac{1}{2}\left[1(8-2) + 0(2-1) + 2(1-8)\right]\right| = 4$ ..... (2)

$b^2 = 2$, $a = 2b = 2\sqrt{2}$ length of latus rectum $= \frac{2b^2}{a} = \frac{4}{2\sqrt{2}} = \sqrt{2}$

41. Required probability $= \frac{m}{m+n} \times \frac{m+5}{m+n+5} + \frac{n}{m+n} \times \frac{m}{m+n+5} = \frac{1}{5}$

$\frac{m(m+5)+mn}{(m+n)(m+n+5)} = \frac{1}{5}$

$4m^2 + 20m + 3mn = n^2 + 5n$

$m = 2 \Rightarrow n = 8$

$m = 3 \Rightarrow n = 12$

$m = 4 \Rightarrow n = 16$

42. As tangents are perpendicular to each other and its points of intersections $P\left(\frac{7}{5}, -\frac{4}{5}\right)$ lies on directrix. AB is focal chord

Equation of AB is $x - y = 4$, as focus is (α, β) so $\alpha - \beta = 4 \Rightarrow \beta = \alpha - 4$

Now $\angle ASP = 90^o$

$\Rightarrow \frac{-2-\alpha+4}{2-\alpha} \cdot \frac{-\frac{4}{5}-\alpha+4}{\frac{7}{5}-\alpha} = -1$

$\Rightarrow \frac{2-\alpha}{2-\alpha} \cdot \frac{\frac{16}{5}-\alpha}{\frac{7}{5}-\alpha} = -1$

$\Rightarrow \frac{16}{5} - \alpha = \alpha - \frac{7}{5}; \ \alpha = \frac{23}{10}$

$\Rightarrow$ Focus is $\left(\frac{23}{10}, -\frac{17}{10}\right)$

Also, $\ell_1 = AS = \frac{3\sqrt{2}}{10}$, $\ell_2 = BS = \frac{27\sqrt{2}}{10}$

$\frac{1}{a} = \frac{1}{\ell_1} + \frac{1}{\ell_2} \Rightarrow a = \frac{27\sqrt{2}}{100} \Rightarrow 4a = \frac{27\sqrt{2}}{25}$

43. Let plane containing line $3x - y + 2z - 1 = 0 = x + 2y - z - 2$ is

$3x - y + 2z - 1 + \lambda(x + 2y - z - 2) = 0$ ..... (1)

(1) must be perpendicular to $3x + 2y + z = 0$ ..... (2)

So that (1) and (2) will give required projected line

$(3 + \lambda)3 + (2\lambda - 1)2 + (2 - \lambda)1 = 0$

$9 + 3\lambda + 4\lambda - 2 + 2 - \lambda = 0$

$6\lambda + 9 = 0$

$\lambda = -\frac{3}{2}$

(1) becomes $3x - y + 2z - 1 - \frac{3}{2}(x + 2y - z - 2) = 0$

$6x - 2y + 4z - 2 - 3x - 6y + 3z + 6 = 0$

$3x - 8y + 7z + 4 = 0$

So, projected line $3x - 8y + 7z + 4 = 0 = 3x + 2y + z$

Also, its symmetrical form $\frac{x+1}{11} = \frac{y-1}{-9} = \frac{z-1}{-15}$

44. (I) Locus of centre of circle S is parabola

(iv) Put $z = x + iy$

$\sqrt{(x-1)^2 + (y-2)^2} = |x - 4 + 1|$

$x^2 - 2x + 1 + y^2 - 4y + 4 = x^2 - 6x + 9$

$\Rightarrow (y - 2)^2 = -4x + 8$ represents parabola

45. (II) S touches one circle externally and other circle internally so centre of S lies on ellipse. So locus is ellipse

(iii) $|z - 1 - i| + |z - 1 - 2i| = 4$

Represents ellipse whose foci are (1, 1), (1, 2) and major axis length $2a = 4$

46. (III) S touches given two isolated circle externally so locus of centre of S is hyperbola

(ii) $||z| - |z - 4 - 5i|| = 3$ represents hyperbola whose foci are (0, 0), (4, 5) and transverse axis length $(2a) = 3$

(IV) Represents line

(i) Represents line

47.-49. (I) $(\vec{a} + 2\vec{b} - \vec{c}) \cdot (\vec{a} - \vec{b}) \times (\vec{a} - \vec{b} - \vec{c}) = 3[\vec{a}\ \vec{b}\ \vec{c}] = 6$

(II) Let $I = \int_0^1 \frac{\tan^{-1} x}{\cot^{-1}(1 - x + x^2)} dx = \int_0^1 \frac{\tan^{-1} x}{\tan^{-1} x - \tan^{-1}(x-1)} dx = \int_0^1 \frac{\tan^{-1}(1-x)}{\tan^{-1}(1-x) + \tan^{-1} x} dx$

$\Rightarrow I = \frac{1}{2}$

(III) Range of $f(x) = \sqrt{x^2 + 5x + 8} - \sqrt{x^2 - 3x + 4}$ is (–4, 4)

(IV) True

(ii) Number of solution of $2\cos x - |\sin x| = 0$ in $[0, 4\pi]$ is 4

(iv) Number of solution of $2^{\cos x} = |\sin x|$ is $[0, 2\pi]$ is 4

(P) is false

(Q) is true

(R) is true

(S) is false, correct answer $\frac{a}{L} = 2$

SECTION – C

50. $\sum_{k=1}^{2017} \left(k^2 + k - (k-1)\right) \lfloor k$

$= \sum_{k=1}^{2017} \left(k \lfloor k+1 - (k-1) \lfloor k\right)$

$= (1 \cdot \lfloor 2 - 0) + (2\lfloor 3 - 1 \cdot \lfloor 2) + (3\lfloor 4 - 2\lfloor 3) + ...(2017\lfloor 2018 - 2016\lfloor 2017)$

$= 2017|\underline{2018} = |\underline{2019} - 2|\underline{2018}$

$a = 2019; b = 2; c = 2018$

$a + b - c = 3$

51. $(1 + 2x + 3x^2 + 4x^3 + ... \infty)^2$

$= \frac{1}{(1-x)^4} = (1-x)^{-4} = 1 + {}^4C_1x + {}^5C_2x^2 + {}^6C_3x^3 + {}^7C_4x^4 + ...$

Coefficient of $x^4 = {}^7C_4 = 35 = m$

Number of divisors of m is 4

52. $(2x)^2 = 12 - 4(x^2 + y^2)$

$8x^2 + 4y^2 = 12$

$\frac{x^2}{3/2} + \frac{y^2}{3} = 1$

$x = \sqrt{\frac{3}{2}}\cos\theta ;\ y = \sqrt{3}\sin\theta$

$2Re(z) + Im(z) = 2x + y = \sqrt{6}\cos\theta + \sqrt{3}\sin\theta \le 3$

53. Given equation can be written as $(x - 1)(x^2 - x \sin x - \cos x) = 0$

Let $f(x) = x^2 - x \sin x - \cos x$

$f'(x) = x(2 - \cos x) = 0; x = 0$

$f(0) = -1$

$f\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4} - \frac{\pi}{2} = \frac{\pi^2 - 2\pi}{4} > 0 ;\ f\left(-\frac{\pi}{2}\right) = \frac{\pi^2}{4} - \frac{\pi}{2} = \frac{\pi^2 - 2\pi}{4} > 0$

Total number of roots in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ is 3

54. Let $P = \begin{bmatrix} \alpha_1 & \alpha_2 \\ \beta_1 & \beta_2 \end{bmatrix}$

Now $P\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \Rightarrow \begin{bmatrix} \alpha_1 & \alpha_2 \\ \beta_1 & \beta_2 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

$\Rightarrow \alpha_1 - \alpha_2 = -1$ ..... (1)

$\Rightarrow \beta_1 - \beta_2 = 2$ ..... (2)

Also, $P^2\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\Rightarrow P\left(P\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow P\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\Rightarrow -\alpha_1 + 2\alpha_2 = 1$ ..... (3)

$\Rightarrow -\beta_1 + 2\beta_2 = 0$ ..... (4)

From equation (1), (2), (3) and (4)

$\alpha_1 = -1; \alpha_2 = 0; \beta_1 = 4; \beta_2 = 2$

Now $\det(P - xI) = 0$

$\Rightarrow \det\left(\begin{bmatrix} -1 & 0 \\ 4 & 2 \end{bmatrix} - x\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 0 \Rightarrow \begin{vmatrix} -1-x & 0 \\ 4 & 2-x \end{vmatrix} = 0$

$\Rightarrow x = 2, -1$

$x_1^2 + x_2^2 = 4 + 1 = 5$